

WILSON SCIENCE QUIZ 2083

SENIOR Division

Official Questions & Answers

RULES OF PLAY

RULES OF PLAY

- Standard rounds (Biology, Physics, Chemistry, Geology & Astronomy, Miscellaneous): 30 seconds per question.
- Audio Visual round: 10 seconds per question (Q1-Q8). Q9 is special: the video plays for 40 seconds, then 40 seconds are given to answer.
- Gambling round (Gold: 5 marks, Diamond: 10 marks, Platinum: 15 marks - 4 questions each): 30 seconds per question.
- Rapid Fire (Sets A-D, 15 questions each): 1 minute per set.

ROUNDS AT A GLANCE

Round	Format	Time allowed
Biology	8 questions	30 seconds each
Physics	8 questions	30 seconds each
Chemistry	8 questions	30 seconds each
Geology and Astronomy	8 questions	30 seconds each
Miscellaneous	8 questions	30 seconds each
Audio Visual	9 questions	10 sec each (Q9 special)
Rapid Fire	4 sets × 15 questions	1 minute per set
Gambling	3 tiers × 4 questions	30 seconds each

Total questions: 121

ROUND 1

BIOLOGY

8 questions · 30 seconds each

Q1. Which blood vessel carries oxygenated blood from lungs to the heart?

Answer: *Pulmonary vein*

Q2. Which brain chemical is often nicknamed as the 'feel good' chemical?

Answer: *Dopamine*

Q3. Which is the largest internal organ in the human body?

Answer: *Liver*

Q4. What is the name of the process in which plants release water vapour into the air?

Answer: *Transpiration*

Q5. Where are RBC formed?

Answer: *In bone marrow*

Q6. Animals of which phylum contains hollow body cavity?

Answer: *Coelenterata*

Q7. Who first crystallized the virus?

Answer: *Stanley*

Q8. Which cell organelle is called traffic police of the cell?

Answer: *Golgi body*

ROUND 2

PHYSICS

8 questions · 30 seconds each

Q1. Which particle mainly carries electric current through a metal wire?

Answer: *Electron*

Q2. Hot water of mass 10 kg at 100°C is mixed with 20 kg water at 10°C. What is the final temperature of water?

Answer: *40°C*

Q3. The time period of a simple pendulum is 2 seconds. Find its frequency.

Answer: *0.5 Hz*

Q4. Which principle says that up thrust is equal to the weight of the displaced liquid?

Answer: *Archimedes' Principle*

Q5. What type of wire is indicated by blue or black colour in household electrical wiring?

Answer: *Neutral wire*

Q6. What is the value of angle of dip at the north pole?

Answer: *90°*

Q7. What is the pressure exerted by the water at the depth of 20 m assuming acceleration due to gravity is 10 m/s²?

Answer: *200000 Pa*

Q8. What happens to the mass of a body when it is taken to the moon?

Answer: *Remains same*

ROUND 3

CHEMISTRY

8 questions · 30 seconds each

Q1. What is the full form of pH?

Answer: *Potential of Hydrogen or Power of Hydrogen*

Q2. How heavier is one proton compared to one electron?

Answer: *1839 times*

Q3. What is the symbol of tin?

Answer: *Sn*

Q4. What happens when an alkaline solution is treated with phenolphthalein?

Answer: *Turns into pink colour*

Q5. Which gas is produced when vinegar reacts with baking soda?

Answer: *CO₂*

Q6. What is the molecular formula of caustic soda?

Answer: *NaOH*

Q7. What is the only non-metal that is liquid at room temperature?

Answer: *Bromine*

Q8. If an atom loses an electron, what does it become?

Answer: *A cation*

ROUND 4

GEOLOGY AND ASTRONOMY

8 questions · 30 seconds each

Q1. What is the first plant sown on the moon?

Answer: *Cotton*

Q2. What is the most abundant mineral found in the earth's crust?

Answer: *Feldspar*

Q3. Dinosaurs are the evolutionary ancestors of what modern day animals?

Answer: *Birds*

Q4. In which planet days are longer than years?

Answer: *Venus*

Q5. What is the point directly above an earthquake's focus called?

Answer: *Epicenter*

Q6. Which is the shortest night in a year?

Answer: *21st June*

Q7. What is the weight of NepaliSat 1?

Answer: *1.3 Kg*

Q8. If the earth suddenly stops rotating, which direction would objects tend to move?

Answer: *Eastward (due to inertia of motion)*

ROUND 5

MISCELLANEOUS

8 questions · 30 seconds each

Q1. How much is one astronomical unit?

Answer: $1.5 \times 10^{11} \text{ m}$ OR 15 crore Km

Q2. What is the collection of related files stored together called?

Answer: Folder

Q3. What is the main function of ribosome?

Answer: Protein synthesis

Q4. Which element has the highest electrical conductivity at room temperature?

Answer: Silver (Ag)

Q5. Which memory loses its data when power is switched off?

Answer: RAM

Q6. What is the main greenhouse gas responsible for the natural greenhouse effect on the Earth?

Answer: Water vapour

Q7. Which hormone helps kidney to reabsorb water back from urine into blood stream?

Answer: Vasopressin or anti-diuretic hormone

Q8. Which division of the kingdom plantae includes moss and Riccia?

Answer: Bryophyta

ROUND 6

AUDIO VISUAL

9 questions · 10 sec each (Q9 special)

Q1. Identify the picture. [IMAGE: Senior audio visual quiz image]

Answer: *Conical Flask*

Q2. Which two reactants are used in this fire extinguisher? [IMAGE: Red fire extinguisher with hose and handle]

Answer: NaHCO_3 and H_2SO_4

Q3. Which method of artificial vegetative propagation is shown in the video? [VIDEO]

Answer: *Grafting*

Q4. Identify the cause of this event. [IMAGE: Senior audio visual quiz image]

Answer: *Anomalous expansion of water*

Q5. What microscopic observation is shown in the picture? [IMAGE: Senior audio visual quiz image]

Answer: *TS of gills of mushroom*

Q6. Identify the logo. [IMAGE: Senior audio visual quiz image]

Answer: *Wikipedia*

Q7. Identify the experiment. [IMAGE: Senior audio visual quiz image]

Answer: *Ammonia gas preparation*

Q8. Identify the famous Robot. [IMAGE: Senior audio visual quiz image]

Answer: *Sophia*

Q9. Last question for Audience: According to the lyrics, which element is mentioned for your pretty teeth? [VIDEO]

Answer: *Fluorine*

ROUND 7

RAPID FIRE

4 sets × 15 questions · 1 minute per set

SET A — 1 MINUTE

Q1. What is the neutron number of silicon?

Answer: *14*

Q2. What happens when ammonia reacts with carbon dioxide?

Answer: *Urea formation*

Q3. What is the SI unit of electric charge?

Answer: *Coulomb*

Q4. What is the full form of VPN?

Answer: *Virtual Private Network*

Q5. Who discovered electromagnetic induction?

Answer: *Faraday*

Q6. What is the scientific name of dog?

Answer: *Canis familiaris*

Q7. What is the phylum of sycon?

Answer: *Porifera*

Q8. What happens to the resistance of a metal conductor when its temperature increases?

Answer: *Increases*

Q9. Which hormone causes aging in plants?

Answer: *Abscisic acid*

Q10. What type of bond is formed by transfer of electrons?

Answer: *Ionic/covalent*

Q11. What is the symbol of iron?

Answer: *Fe*

Q12. What is the study of honey bee called?

Answer: *Apiology*

Q13. 1 m is equal to how many millimetres?

Answer: 1000

Q14. Which vitamin deficiency causes night blindness?

Answer: Vitamin A

Q15. Which planet has ice caps on both of the poles?

Answer: Mars

SET B — 1 MINUTE

Q1. What is the neutron number of Sulphur?

Answer: 16

Q2. What happens when carbon dioxide reacts with lime water for short period of time?

Answer: CaCO_3 formation

Q3. What is the SI unit of specific heat capacity?

Answer: $\text{J}/(\text{kg} \cdot ^\circ\text{C})$

Q4. What is the full form of HTML?

Answer: Hypertext markup language

Q5. Which device is used to measure the density of milk?

Answer: Lactometer

Q6. What is the scientific name of rice?

Answer: *Oryza sativa*

Q7. What is the phylum of pinworm?

Answer: *Nemathelminthes*

Q8. What is the weight of human brain?

Answer: 1.35 Kg

Q9. Why does a person feel lighter in an elevator?

Answer: Accelerating downwards

Q10. Which is the strongest evidence of evolution?

Answer: Fossils

Q11. What is the symbol of zinc?

Answer: *Zn*

Q12. What is the study of diseases called?

Answer: *Pathology*

Q13. How many metres is equal to 1 micrometre?

Answer: *1/10⁶ m*

Q14. Rickets is caused by which vitamin deficiency?

Answer: *Vitamin D*

Q15. What makes Mars look red in colour?

Answer: *Iron oxide*

SET C — 1 MINUTE

Q1. What is the valency of phosphorus?

Answer: *3*

Q2. What happens when ammonia reacts with water?

Answer: *Forms ammonium hydroxide*

Q3. What is the SI unit of gravitational constant?

Answer: *N·m²/kg²*

Q4. What is the full form of HTTP?

Answer: *Hypertext transfer protocol*

Q5. What is the value of absolute zero in degree Celsius?

Answer: *-273.15°C*

Q6. What is the scientific name of potato?

Answer: *Solanum tuberosum*

Q7. What is the phylum of nereis?

Answer: *Annelida*

Q8. Which hormone helps in elongation of stem in plants?

Answer: *Gibberellin*

Q9. What happens to the boiling point of water at higher altitude?

Answer: *Decreases*

Q10. What does CT stand for in medical imaging?

Answer: *Computed tomography*

Q11. What is the symbol of manganese?

Answer: *Mn*

Q12. What is the study of bird called?

Answer: *Ornithology*

Q13. How many metres is one nanometre?

Answer: *1/10⁹ m*

Q14. Marasmus is caused by which nutrition deficiency?

Answer: *Carbohydrates*

Q15. Which planet is lighter than water?

Answer: *Saturn*

SET D — 1 MINUTE

Q1. What is the atomic mass of aluminum?

Answer: *27 amu*

Q2. What happens when a burning magnesium ribbon is exposed to carbon dioxide?

Answer: *MgO forms with carbon patches*

Q3. What is the SI unit of luminous intensity?

Answer: *Candela*

Q4. What is the full form of SSD?

Answer: *Solid State Drive*

Q5. Which device measures liquid pressure?

Answer: *Pressure gauge / manometer*

Q6. What is the scientific name of cow?

Answer: *Bos taurus*

Q7. What is the phylum of unio?

Answer: *Mollusca*

Q8. What is the weight of pituitary gland?

Answer: *1.5 grams*

Q9. What happens when an object cannot displace water equal to its weight?

Answer: *Object will sink*

Q10. What is called the hybrid made from male horse and female donkey?

Answer: *Hinny*

Q11. What is the symbol of iodine?

Answer: *I*

Q12. What is the study of fossils called?

Answer: *Paleontology*

Q13. What value is represented by prefix deca?

Answer: *10*

Q14. Goitre is caused by which hormone deficiency?

Answer: *Thyroxine*

Q15. Which is the coldest planet in solar system?

Answer: *Uranus*

ROUND 8

GAMBLING

3 tiers × 4 questions · 30 seconds each

GOLD — 5 MARKS PER QUESTION

Q1. What is the SI unit of resistance?

Answer: *Ohm*

Q2. What is the molecular formula of quick lime?

Answer: *CaO*

Q3. Which part of brain balances body movement and postures?

Answer: *Cerebellum*

Q4. What is the process called when solid directly changes into gas?

Answer: *Sublimation*

DIAMOND — 10 MARKS PER QUESTION

Q1. Which material is used to shield against radiation?

Answer: *Lead (Pb)*

Q2. Which human organ uses the most oxygen?

Answer: *Brain*

Q3. What is the scientific name of titepati?

Answer: *Artemisia vulgaris*

Q4. What is the escape velocity of the Earth?

Answer: *11.2 km/s*

PLATINUM — 15 MARKS PER QUESTION

Q1. Calculate the molecular weight of slaked lime.

Answer: *74 g/mol or 74 amu*

Q2. What is the name of the satellite developed by Nepali secondary school students that was launched in January 2026?

Answer: *Munal*

Q3. What is the focal length of plane mirror?

Answer: *Infinity*

Q4. Which scientist proposed the heliocentric model of the solar system?

Answer: *Nicolaus Copernicus*